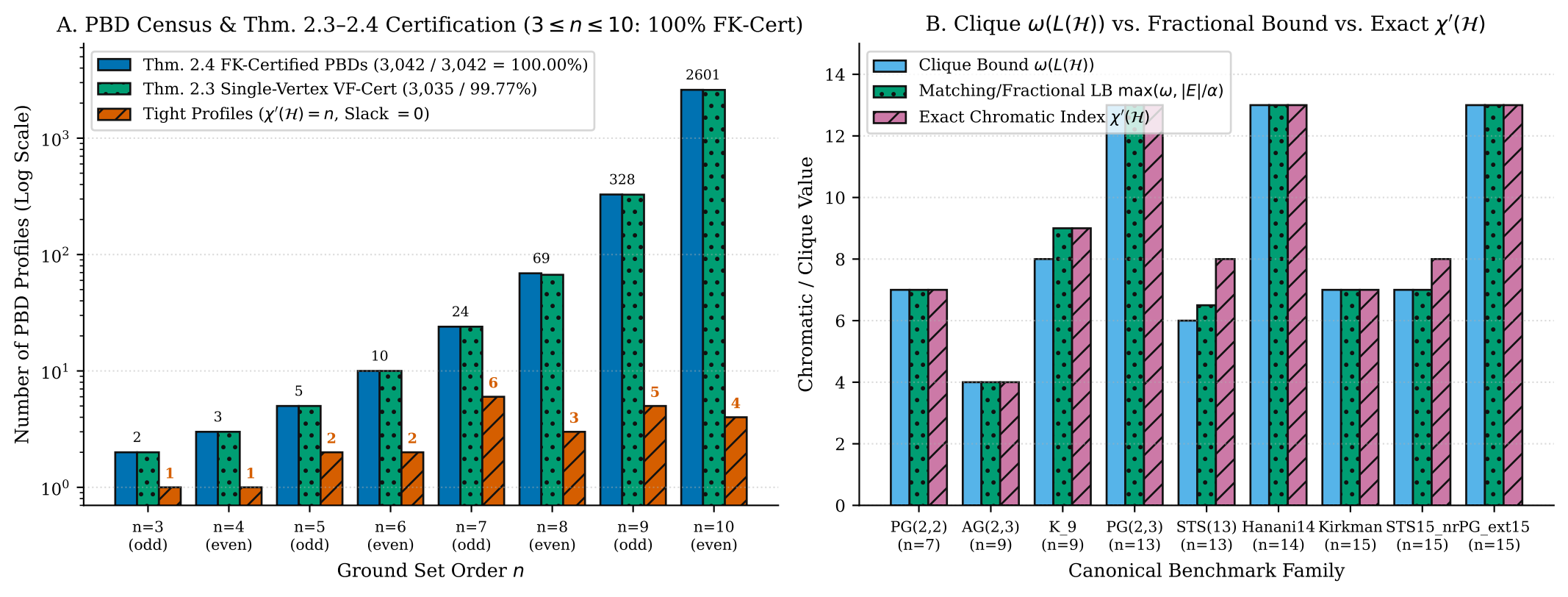

A. PBD Census & Thm. 2.3–2.4 Certification ($3 \le n \le 10$: 100% FK-Cert)
Thm. 2.4 FK-Certified PBDs (3,042 / 3,042 = 100.00%)
Thm. 2.3 Single-Vertex VF-Cert (3,035 / 99.77%)
Tight Profiles ($\chi'(\mathcal{H}) = n$, Slack $= 0$)
Number of PBD Profiles (Log Scale)
10^0
10^1
10^2
10^3
2
1
3
1
5
2
10
2
24
6
69
3
328
5
2601
4
n=3 (odd)
n=4 (even)
n=5 (odd)
n=6 (even)
n=7 (odd)
n=8 (even)
n=9 (odd)
n=10 (even)
Ground Set Order n
B. Clique $\omega(L(\mathcal{H}))$ vs. Fractional Bound vs. Exact $\chi'(\mathcal{H})$
Clique Bound $\omega(L(\mathcal{H}))$
Matching/Fractional LB $\max(\omega, |E|/\alpha)$
Exact Chromatic Index $\chi'(\mathcal{H})$
Chromatic / Clique Value
0
2
4
6
8
10
12
14
PG(2,2) (n=7)
AG(2,3) (n=9)
K_9 (n=9)
PG(2,3) (n=13)
STS(13) (n=13)
Hanani14 (n=14)
Kirkman (n=15)
STS15_nr (n=15)
PG_ext15 (n=15)
Canonical Benchmark Family